

# INJIN KONG

## Curriculum Vitae

### Personal Information

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|---------------------|------------------------------------------------------------|
| AFFILIATION         | Graduate School of Data Science, Seoul National University |
| PHONE               | +82-10-6243-2892                                           |
| EMAIL               | <a href="mailto:mtkong77@gmail.com">mtkong77@gmail.com</a> |
| ADVISOR             | Yohan Jo                                                   |
| BIRTH / NATIONALITY | August 24, 2000 / Republic of Korea                        |
| LANGUAGE            | Native Korean, fluent in English                           |

### Education

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|-----------|-------------------------------------------------------------------------------------|--------------|
| CURRENT   | <b>Seoul National University</b><br>M.S. in Data Science<br>Advisor: Prof. Yohan Jo | Seoul, Korea |
| FEB. 2025 | <b>Seoul National University</b><br>B.S. in Mathematics                             | Seoul, Korea |
| FEB. 2019 | <b>Hana Academy Seoul</b>                                                           | Seoul, Korea |

### Research Interests

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My research focuses on the **inner workings of language models**, with a particular interest in applying **diffusion-based generative modeling** to language generation. I study how language models represent information, organize internal computation, and support reliable applications in reasoning, agents, and education.

- Inner workings of neural language models
- Mechanistic interpretability and representation analysis
- Diffusion-based and autoregressive language generation
- Language model applications, including reasoning, agents, and education

### Publications

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\*Equal contribution.

#### Peer-Reviewed Publications

- [1] Jongwook Han\*, Jongwon Lim\*, **Injin Kong**, and Yohan Jo. “Dual Mechanisms of Value Expression: Intrinsic vs. Prompted Values in Large Language Models.” *International Conference on Machine Learning (ICML)*, 2026.

#### Manuscripts Under Review

- [1] **Injin Kong**, Hyoungjoon Lee, and Yohan Jo. “Where Should Diffusion Enter a Language Model? Geometry-Guided Hidden-State Replacement.” *arXiv preprint arXiv:2605.14368*, 2026.
- [2] Naeun Lee, Hyunjong Kim, Sunghwan Choi, **Injin Kong**, and Yohan Jo. “Can MLLMs Reason About Visual Persuasion? Evaluating the Efficacy and Faithfulness of Reasoning.” *arXiv preprint arXiv:2605.08965*, 2026.
- [3] **Injin Kong**\*, Hyoungjoon Lee\*, and Yohan Jo. “Mechanism Shift During Post-training from Autoregressive to Masked Diffusion Language Models.” *arXiv preprint arXiv:2601.14758*, 2026.

- [4] **Injin Kong**, Shinyee Kang, Yuna Park, Sooyong Kim, and Sanghyun Park. “Style Extraction on Text Embeddings Using VAE and Parallel Dataset.” *arXiv preprint arXiv:2502.08668*, 2025.

## Research Experience

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### Graduate Researcher, Seoul National University

Mar. 2025 – Present

HOLI Lab, advised by Prof. Yohan Jo

- Conduct research on language models, diffusion-based generation, mechanistic interpretability, and multimodal reasoning.
- Develop mathematical formulations and theoretical frameworks for analyzing generation mechanisms, decoding dynamics, and representation-level behavior.
- Analyze internal representations and mechanism shifts in language models, especially across autoregressive and diffusion-based generation.

### Chief Scientific Officer, Aardvark

2023 – 2025

Aardvark Research Team

- Led research on style extraction from text embeddings and AI-based educational content generation.
- Developed a VAE-based approach for analyzing stylistic variation in text embeddings using parallel text data.

## Entrepreneurial & Industry Experience

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### Aardvark

2023 – 2025

Co-Founder and Chief Scientific Officer, AI EdTech Startup

- Built AI systems for generating English test questions, including data construction, retrieval, model prompting, and evaluation pipelines.
- Led research and product development for LLM-based educational content generation.

### BLACKLABEL Geometry

2020

Mathematics Content Reviewer, Jinhaksa

- Reviewed advanced geometry problems for high-achieving students in *BLACKLABEL Geometry*, a mathematics workbook based on the 2015 revised Korean national curriculum.

## Awards & Honors

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|------|-----------------------------------------------------------------------------------------------------------------------|
| 2023 | CJ Logistics Future Technology Challenge, <b>Second Place</b><br><i>Image-based parcel volume measurement project</i> |
| 2019 | Yonsei University, <b>Academic Excellence Award</b><br><i>Highest academic achievement</i>                            |
| 2018 | Korean Mathematical Olympiad (KMO), <b>Gold Prize</b>                                                                 |
| 2018 | Hana Academy Seoul Earth Science Competition, <b>Grand Prize</b><br><i>1st place</i>                                  |
| 2018 | Hana Academy Seoul Mathematics Research Presentation Contest, <b>Gold Prize</b><br><i>2nd place</i>                   |

## Academic Activities

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|---------------------------|---------------------------------------------|------|
| <b>Reviewer</b>           | ICML 2026 Workshop on Pluralistic Alignment | 2026 |
| <b>Teaching Assistant</b> | Computing 1, Seoul National University      | 2025 |

## Skills

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**Programming and Tools.** Python, PyTorch, Hugging Face Transformers, L<sup>A</sup>T<sub>E</sub>X, Git, Linux

**Research Skills.** Mechanistic interpretability, diffusion language modeling, representation analysis, hidden-state geometry, multimodal rationale evaluation